Exploring the Relationship Between Household Income and Happiness in the UK: Data from the European Social Survey (ESS)

Words count:1247(Without references)

1287(With references)

1 Aims

Currently, the global economy is in a downturn and a large amount of people in the UK are suffering from the cost of living crisis. According to the data (Office for National Statistics (ONS), 2025), the inflation rate has climbed 3.7%. However, the rate of income increase may not catch the inflation rate, individuals may feel stressed and unhappier. This aroused my interest in the Research Question: Are individuals in the UK who have higher household incomes happier?

The hypothesis test I want to test is

- Null Hypothesis (H_0): There is no relationship between household income and happiness among individuals in the UK.

- Alternative Hypothesis (H_A): Individuals in the UK with higher household incomes report higher levels of happiness.

2 Measures

Variable Type	Variable Name	Description	measurement type	Mean	Median	Min	Max	Std.Dev	Recording
dependent	**happy**	**Overall happiness score (0–10)**	**Continuous**	**7.54**	**8.0**	**2**	**10**	**1.82**	**No**
independent	**hinctnta**	**Household income decile (1–10)**	**Ordinal**	**5.29**	**5.0**	**1**	**10**	**2.98**	**Handle missing values (NA)**
Control	**eisced**	**Education level (0–6)**	**Ordinal**	**-**	**-**	**-**	**-**	**-**	**Recode into 3 categories (edu_3cat)** **Low**: 0–2 **Middle**: 3–4 **High**: 5–6
Control	**health**	**Self-rated health (1 = very good, 5 = very bad)**	**Continuous**	**2.21**	**2.0**	**1**	**5**	**0.96**	**Needs reverse coding**
Control	**gndr**	**Gender (1 = Male, 2 = Female)**	**Nominal**	**-**	**-**	**-**	**-**	**-**	**Recode to binary variable (0 = M, 1 = F)**
Control	**alcfreq**	**Drink frequency (1=everyday, 7=never)**	**Continuous**	**4.17**	**4.0**	**1**	**7**	**2.06**	**Needs reverse coding**

Note: Distribution of categorical variables

```
  Variables       Category     N      %
  <chr>           <chr>    <int>    <dbl>
1 Education Level Low        532     31.6
2 Education Level Middle     347     20.6
3 Education Level High       520     30.9
4 Gender          Men        824     48.9
5 Gender          Women      860     51.1
---------------------------------------------
```

Before embarking on any analytical work, it is necessary to identify and treat outliers.
(Please see the R-code for details 2.1-2.5)

3 Visualisation

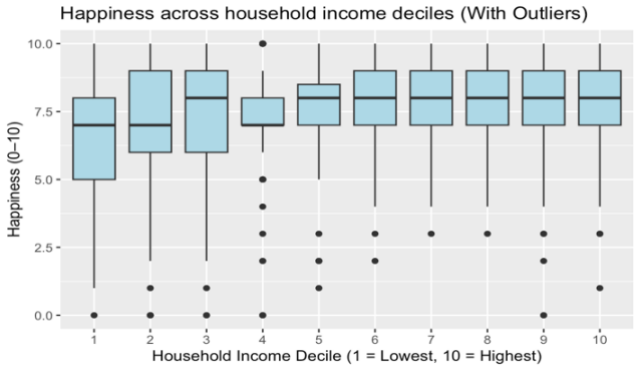

Figure 1

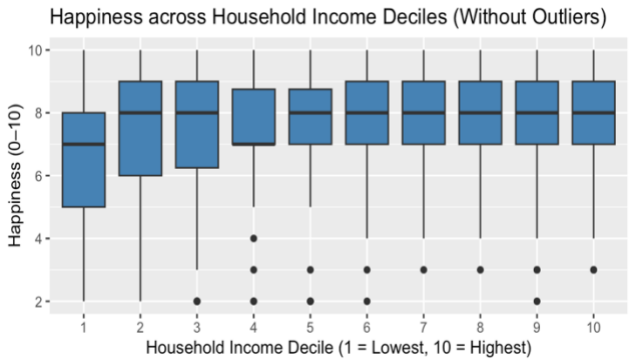

Figure 2

Figure 1 shows the distribution of self-reported happiness across ten household income deciles(with outliers,indicating a positive strong correlation between happiness and household income decile, as the income increases, median happiness scores also tend to rise. Additionally, the figure 2(without outliers) provides a more steady upward trajectory between these two variables.

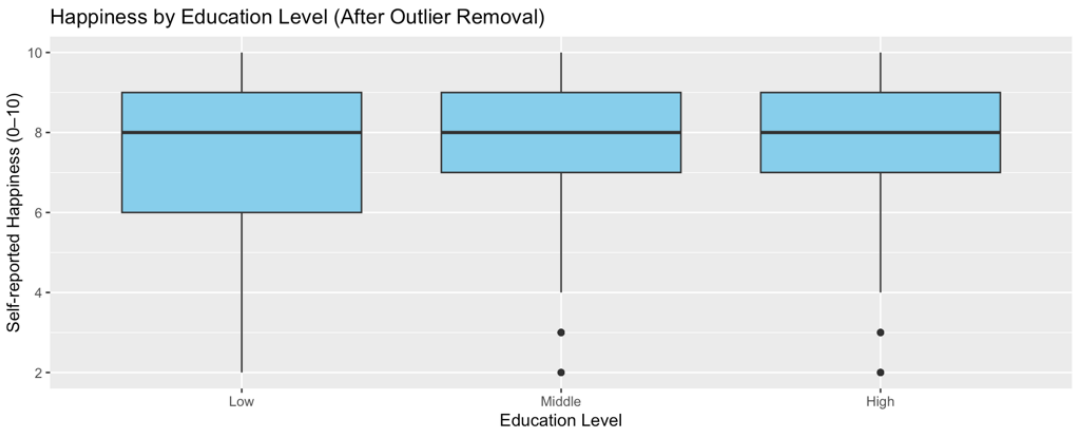

Figure 3

4 Own visualisation

Most individuals may assume that with a higher education degree, they may feel happier. The **figure 3** shows happiness levels across three grouped categories of educational level(without outliers): low, middle, and high. All three of the groups' median happiness scores stay constant at approximately 8. However, even at the higher education level, there are also outliers in the low happiness. This suggests that education alone may not be a strong determinant of happiness; other factors like health, gender, drinking frequency (especially in the UK) may still be considerable factors.

5. Regression

Before doing the regression, it still needs to do some housekeeping. (Please see code 5.1)
To investigate the relationship between these variables, one regression model was fitted. The outcome of model 1, which includes household income as the primary predictor, controls for self-rated health, gender, alcohol consumption frequency, and education level, is shown in Table 1.

Table 1. Linear Regression Model Predicting Happiness from Household Income and Control Variables

(Intercept)	9.262***
	(0.327)
hinctnta(Household income)	0.026
	(0.023)
health	-0.797***
	(0.064)
Gndr(Female)	0.035
	(0.121)
alcfreq	-0.039
	(0.030)
edu_3catMiddle	0.072
	(0.161)
edu_3catHigh	-0.181
	(0.150)
Num.Obs.	988
R2	0.160
R2 Adj.	0.155

Note: Reference categories: education(low), Gndr(Female),
Significance levels: p < 0.1, * p < 0.05, p < 0.01, * p < 0.001

The regression model 1 partially support H_0, shows a weak positive relationship between happiness and household income decile **(hinctnta)**. For every 1-unit increase in hinctnta, happiness is expected to increase by 0.026 (β =0.026 , SE = 0.023, p = 0.229).
In contrast, **self-rated health (reverse coded)** suggests a strong relationship (β =-0.797, SE = 0.064, p = < 2e-16). Indeed, low happiness can raise the risk of being unhealthy (Steptoe, 2019).
Other control variables, including **gndrFemale** (β = 0.035, *p* = 0.956), **drinking frequency** (alcfreq_rev: β = -0.039, *p* = 0.130), and **education level** (edu_3catMiddle: β = 0.072, *p* = 0.690; edu_3catHigh: β = –0.181, *p* = 0.109), were **not statistically significant**.

In conlusion, based on 988 observations, the model accounts for about 16% of the variation in happiness (Adjusted R2 = 0.155). UK citizens with higher income may not feel much happier, but with healthier condictions do. Therefore, the H_0 is not supported by the results. Additionally, to get a more precise result, Further analyses and alternative models are recommended.

6 Appendix

```
## Load necessary packages ----
pacman::p_load(modelsummary, tidyverse, psych, haven)
## Import the data
data <- read_dta("ESS11-subset.dta")
table(data$cntry)
glimpse(data)
## 2.Measure
## 2.1 View the data
psych::describe(data_clean[, c("happy", "hinctnta", "health", "gndr", "alcfreq")])

## 2.3 Drop NA
## MCAR missing completely at random - listwide deletion
data_clean <- data %>%
  select(happy, hinctnta, eisced, health, gndr, alcfreq) %>%
  drop_na()
glimpse(data_clean)
## 2.4 Save a version BEFORE outlier removal
data_no_outlier_removal <- data_clean
## 2.5 Visually inspect continuous measure' potential outliers
boxplot_vars <- c("happy", "health", "alcfreq")
boxplot(data_clean[, boxplot_vars])
## Only maintain mean、sd、min、max
outl <- psych::describe(data_clean[, boxplot_vars], fast = TRUE)
outl <- data.frame(outl[, c(3:6)])
bounds <- outl %>%
  mutate(
    lower = mean - 3 * sd,
    upper = mean + 3 * sd)
data_clean <- data_clean %>%
  mutate(
    happy_outl = ifelse(happy > bounds["happy", "upper"] | happy < bounds["happy", "lower"], 1, 0),
    health_outl = ifelse(health > bounds["health", "upper"] | health < bounds["health", "lower"], 1, 0),
    alcfreq_outl = ifelse(alcfreq > bounds["alcfreq", "upper"] | alcfreq < bounds["alcfreq", "lower"], 1,
0)
  )
data_clean <- data_clean %>%
  mutate(outlier = ifelse(happy_outl == 1 | health_outl == 1 | alcfreq_outl == 1, 1, 0))
## Check numbers of outliers
table(data_clean$outlier)
## Delete outliers and keep the "clean data"
data_clean <- data_clean %>% filter(outlier == 0)
## 3 Visualise outcome variable and main predictor
ggplot(data_no_outlier_removal, aes(x = factor(hinctnta), y = happy)) +
  geom_boxplot(fill = "lightblue") +
  labs(
    title = "Happiness across household income deciles (With Outliers)",
    x = "Household Income Decile (1 = Lowest, 10 = Highest)",
    y = "Happiness (0–10)"
  )
```

```r
ggplot(data_clean, aes(x = factor(hinctnta), y = happy)) +
  geom_boxplot(fill = "steelblue") +
  labs(
    title = "Happiness across Household income Deciles (Without Outliers)",
    x = "Household Income Decile (1 = Lowest, 10 = Highest)",
    y = "Happiness (0–10)"
  )

## 4 Own visualisation ----
## Clean data and mutate edu_3cat
data_clean <- data_clean %>%
  mutate(edu_3cat = case_when(
    eisced %in% 0:2 ~ "Low",
    eisced %in% 3:4 ~ "Middle",
    eisced %in% 5:6 ~ "High"
  )) %>%
  mutate(edu_3cat = factor(edu_3cat, levels = c("Low", "Middle", "High")))

ggplot(data_clean %>% filter(!is.na(edu_3cat)),
       aes(x = edu_3cat, y = happy)) +
  geom_boxplot(fill = "skyblue") +
  labs(
    title = "Happiness by Education Level (After Outlier Removal)",
    x = "Education Level",
    y = "Self-reported Happiness (0–10)"
  )
## 5 Regression
## 5.1 # Recode variables(housekeeping)
data_clean <- data_clean %>%
  mutate(
    # Reverse coding
    health_rev = 6 - health,
    alcfreq_rev = 8 - alcfreq,

    # Gender as factor
    gndr = factor(gndr, levels = c(1, 2), labels = c("Male", "Female"))
  )
## 5.2 # Regression
model1 <- lm(happy ~ hinctnta + health_rev + gndr + alcfreq_rev + edu_3cat, data = data_clean)
summary(model1)
modelsummary(model1,
             stars = TRUE,
             gof_map = c("nobs", "r.squared", "adj.r.squared"),
             output = "regression.docx",
             title = "Table 1.Linear Regression Model Predicting Happiness from Household Income and
Control Variables")
```

7 References:

Office for National Statistics (ONS) (2025). *Consumer Price inflation, UK - Office for National Statistics*. [online] www.ons.gov.uk. Available at: https://www.ons.gov.uk/economy/inflationandpriceindices/bulletins/consumerpriceinflation/latest [Accessed 9 Apr. 2025].

Steptoe, A. (2019). Happiness and Health. *Annual Review of Public Health*, [online] 40(1), pp.339–359. doi:https://doi.org/10.1146/annurev-publhealth-040218-044150.